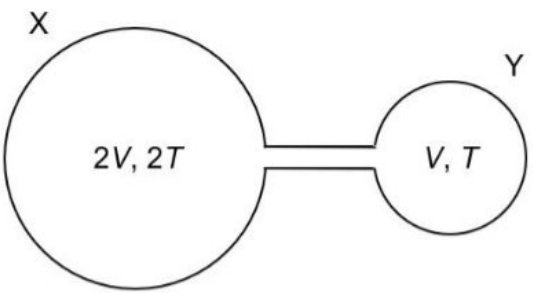
15 An ideal gas is contained in two spherical containers X and Y of volume $2V$ and V respectively, connected by a hollow tube of negligible volume. The containers X and Y are maintained at temperatures $2T$ and T respectively. The setup is shown in the figure below.



Determine the ratio

$$\frac{\text{number of moles of gas in container X}}{\text{number of moles of gas in container Y}}.$$

A 0.25 **B** 1.00 **C** 2.00 **D** 4.00